

▼ Week 0 - Welcome and review**Logistics****Welcome letter****Link to syllabus****Review Materials****Review materials guidelines****[Review] Pseudocode, Python console, PyCharm shortcuts and debugger (Week 4, 6 of 3A)****[Review] OOP (Week 6 and 7 of CS3A)****[Review] Default parameter values, named arguments, exception (Week 9 of CS3A)****[Review] Inheritance, polymorphism; pythonic getter and setter, unittest (Week 4 of CS3B)****[Review] Operator overloading, shallow and deep copy, duck typing (Week 6 of CS3B)****[Review] Function as data, higher-order functions and lambda, mapreduce, filter, iterable (Week 7 of CS3B)**



[Review] *args and **kwargs, MRO, super() (Week 10 of CS3B)

[Supplemental] 3A/3B reviews



Default and Named Parameters, Review of Signatures



Inner Classes and String Methods



A Card Game



The Complete Card Class and Deep Copy Preview



Formatting Text and Numbers



Static Member Data



Static and Class Methods



More About Method Parameters and Class Accessors



Knowledge Check



Design and Implementation of an OOP Project



Completion of the OOP Project

FYI



Student services and tech support



STEM Center intro



Science Learning Institute at Foothill College

Assignments



Intro survey

Apr 11



First Week Introductions

Apr 8



Week 0 discussion

▼ Week 1 - Introduction to data structures and algorithms

Logistics



Link to syllabus



Student-Facing Communication Plan

Class meetings

Class meeting notes and recordings

 0 pts

Lectures

 **Week 1 topics (set, subset sum, itunes)**

[Supplemental] Readings

 **Week 1 readings errata and commentary** **Why We're Here** **Timing is Everything** **Our First Sample Data Set - iTunes *** **Search: The First Main Task** **Inserting and Sorting** **Analysis** **Templates vs. Duck Typing** **An Algorithm** **Getting ready for Subset Sum**

Assignments



Intro survey

Apr 11



First Week Introductions

Apr 8



Reflection on Artificial Intelligence and Computer Science Education

Apr 10 20 pts



Week 1 and Assignment 1 discussion



General Assignment Requirements



Assignment 1 - The Subset-Sum Problem

Apr 13 60 pts



[Optional] Practice Quiz (intro and week 1)



Quiz (intro and week 1)

Apr 17 40 pts



Participation (week 1-6)

May 16 10 pts



Participation extra credit (any time during the quarter)

Jun 26 0 pts



Final project

Jun 24 120 pts



Final project discussion

▼ Week 2 - Generator, Linked Lists, and Sparse Matrices

Lectures



Week 2 topics (iterable and generator, linked list, sparse matrix)



Before-class Quiz 2

Apr 15 5 pts

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings



Week 2 readings errata and commentary



Iterators



Unordered Linked Lists



Adding Accessors and Iterators



Ordered Linked Lists



Sorted Linked Lists



Making an Unordered Object Type Ordered



Sparse Matrices

Assignments



Week 2 and Assignment 2 discussion



Assignment 2 - Sparse Matrices

Apr 20 60 pts

▼ Week 3 - Time Complexity

Lectures



Week 3 topics (big o)



Before-class Quiz 3

Apr 22 5 pts



Big O practice problems

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings



Week 3 readings errata and commentary



Big O



Getting a Tighter Fix on Growth Rates



Big O Estimation from Code



A Few More Examples



Testing the Theory



Analyzing Binary Search



Logarithmic Running Times



Exponential Running Times

Assignments



Week 3 and Assignment 3 discussion



Assignment 3 - Timing Matrix Multiplication

Apr 27 60 pts



Quiz (week 2 and 3)

May 1 40 pts

▼ Week 4 - Binary search tree**Lectures****Week 4 topics (binary search tree, lazy-delete BST)****Before-class Quiz 4**

Apr 29 5 pts

Class meetings**Class meeting notes and recordings**

0 pts

[Supplemental] Readings - General tree**Week 4 readings errata and commentary****General Trees****[Optional] Implementation of Tree Methods****[Optional] Traversals and Function Objects****Deep Copies, Shallow Copies, and Assignment (was covered in 3B, review as necessary)**

[Supplemental] Readings - Binary search tree



Binary Trees



Searching the Binary Search Tree



BST Full Listing



Searching Project Gutenberg *



Analysis and Discussion of BST



Lazy Deletion *

Assignments



Week 4 and Assignment 4 discussion



Assignment 4 - Lazy Deletion in Binary Search Trees

May 4 60 pts



Participation (week 1-6)

May 16 10 pts

▼ Week 5 - Variants of BST

Lectures



Week 5 topics (AVL tree, splay tree)



Before-class Quiz 5

May 6 5 pts

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings - AVL tree



Week 5 readings errata and commentary



AVL Trees



Single Rotation Practice



Double Rotation

[Supplemental] Readings - Top-down splay tree



Top-Down Splaying



Splay Pictures



Splay Algorithms and Closing Tree Remarks

Assignments



Week 5 and Assignment 5 discussion



Assignment 5 - AVL Tree

May 11 60 pts



Quiz (week 4 and 5)

May 15 40 pts

▼ Week 6 - Priority Queues, Binary Heaps, and Min Heaps

Lectures



Week 6 topics (priority queue, min heap)



Before-class Quiz 6

May 13 5 pts

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings



Week 6 readings errata and commentary-2



Priority Queues and Binary Heaps (Min or Max Heaps)

Implementing a Binary Heap



Insert and Percolate Up



Remove and Percolate Down



Accepting a list in __init__()



[Optional] order_heap() Performance

Assignments



Week 6 and Assignment 6 discussion



Assignment 6 - A few min heap coding exercises

60 pts



Participation (week 1-6)

May 16 10 pts



Participation extra credit (any time during the quarter)

Jun 26 0 pts



Mid-quarter survey extra credit claim

May 23 0 pts



Final project

Jun 24 120 pts

▼ Week 7 - Hash Tables

Lectures



Week 7 topics (hash table)



Before-class Quiz 7

May 20 5 pts

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings



Week 7 readings errata and commentary



Hashing Fundamentals



Hashing Functions



Implementing Separate Chaining



Open Addressing



Quadratic Probing



Implementing Quadratic Probing



find() in Hash Tables



Searching Project Gutenberg *

Assignments



Week 7 and Assignment 7 discussion



Assignment 7 - Implementing HashMap

May 25 60 pts



Quiz (week 6 and 7)

May 29 40 pts



Participation (week 7-12)

Jun 26 10 pts



Mid-quarter survey extra credit claim

May 23 0 pts

▼ Week 8 - Insertion, Shell, and Merge Sort

Lectures



Week 8 topics (insertion, shell, and merge sort)



Before-class Quiz 8

May 27 5 pts

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings



Week 8 readings errata and commentary



Insertion Sort



Shell Sort



Improving Shell Sort



MergeSort



Analysis of MergeSort

Assignments



Week 8 and Assignment 8 discussion



Assignment 8 - Analyzing Shellsort's Gaps

Jun 1 60 pts

▼ Week 9 - Heap sort and Quick sort

Lectures



Week 9 topics (heap sort, quick sort)



Before-class Quiz 9

Jun 3 5 pts

Class meetings



Class meeting notes and recordings

0 pts

[Supplemental] Readings



Week 9 readings commentary and errata



Heap Sort



Heapsort, Again



QuickSort



Coding QuickSort (optional)



QuickSort Observations

Assignments



Week 9 and Assignment 9 discussion

**Assignment 9 - Recursion limit and pivot selection of Quick sort**

Jun 8 60 pts

**Quiz (week 8 and 9)**

Jun 12 40 pts

▼ Week 10 - Graph, Dijkstra, max flow**Lectures****Week 10 topics (graph, Dijkstra SPF, max flow)****Before-class Quiz 10**

Jun 10 5 pts

Class meetings**Class meeting notes and recordings**

0 pts

[Supplemental] Readings**Week 10 readings errata and commentary****Motivating the Graph****Graph Data Structures****The Vertex and Adjacency List**



The Edge Class



The Graph Class



The Dijkstra Algorithm



Implementing Dijkstra



The Maximum Flow Problem

Assignments



Week 10 and Assignment 10 discussion



Assignment 10 - The Maximum flow problem

Jun 15 60 pts



Participation (week 7-12)

Jun 26 10 pts



▼ Week 11 - Minimum Spanning Trees, Kruskal's algorithm

Lectures



Week 11 topics (MST, Kruskal)

**Before-class Quiz 11**

Jun 17 5 pts

Class meetings**Class meeting notes and recordings**

0 pts

[Supplemental] Readings**Week 11 readings errata and commentary****Minimum Spanning Trees****Minimum Spanning Trees-2****The Kruskal Algorithm****Analyzing Kruskal****Implementing Kruskal****Full Kruskal Source****Assignments****Week 11 discussion**



Quiz (week 10 and 11)

Jun 26 40 pts

▼ Week 12

Class meetings



Class meeting notes and recordings

0 pts

Assignment and quiz



Final project

Jun 24 120 pts



Participation (week 7-12)

Jun 26 10 pts



Participation extra credit (any time during the quarter)

Jun 26 0 pts

